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1. Purpose

EURODIS offers a restful API in JSON format designed for seamless integration of partner IT systems. This document outlines the logics for data exchange between partners and the central data platform. Priority is given to the document at https://api.eurodis.com, the description of this document is merely an addition.

2. Area of application

This annex is applicable to all EURODIS network partners, as well as any partners involved in sending, receiving, or handling shipments.

3. Links

The description is based on the Swagger infrastructure and can be accessed via the link https://api.eurodis.com/. Additionally, there are both a production environment and a test environment available.

Production environment:

- https://api.eurodis.com/pudo/documentation/index.html
- https://api.eurodis.com/send-pan/documentation/index.html
- https://api.eurodis.com/receive-pan/documentation/index.html
- https://api.eurodis.com/send-tracking/documentation/index.html
- https://api.eurodis.com/receive-tracking/documentation/index.html
- https://api.eurodis.com/send-pod/documentation/index.html
- https://api.eurodis.com/receive-pod/documentation/index.html

Test environment:

- https://testapi.eurodis.com/pudo/documentation/index.html
- https://testapi.eurodis.com/send-pan/documentation/index.html
- https://testapi.eurodis.com/receive-pan/documentation/index.html
- https://testapi.eurodis.com/send-tracking/documentation/index.html
- https://testapi.eurodis.com/receive-tracking/documentation/index.html
- https://testapi.eurodis.com/send-pod/documentation/index.html
- https://testapi.eurodis.com/receive-pod/documentation/index.html

4. Responsibilities

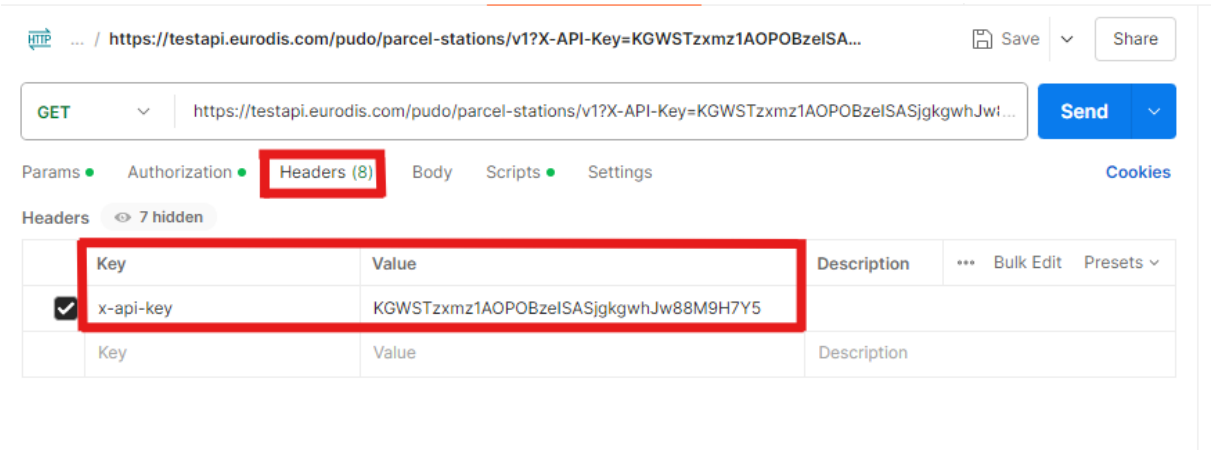
All partners are required to follow the standardized data mapping when exchanging information through the API with the central server, both for sending data from their internal systems and for receiving data.

5. Description

5.1. Authorization

The EURODIS API require the use of an API key for a secure access. They are generated by EURODIS and shared with partners within the integration process.

To access these secure endpoints in the EURODIS test and production environments, you must include a key. Add the API key using the X-API-Key header in your request:



5.2 PAN Exchange (Shipment data)

The endpoint https://api.eurodis.com/send-pan/shipment/v1 is used to create and update a normal shipment or a return within the system. For detailed information on field descriptions, please refer to the provided link.

5.2.1 Create Shipment

To create a shipment, you must follow the field descriptions provided at https://api.eurodis.com/send-pan/documentation/index.html and ensure that the information is placed in the correct fields.

The POST function is used for these purposes:

The ED shipment number and ED collo barcode are generated centrally by EURODIS and returned to the shipping partner within the response. To facilitate this, the fields for ED shipment number and ED collo barcode must be submitted empty in the request by the shipping partner.

Please review the request and response at https://api.eurodis.com/send-pan/documentation/index.html. This clearly shows how they are structured.

5.2.2 Update Shipment

Once the shipment data is submitted to EURODIS via the POST request, it is initially placed in a pre-database. During this time, the data is not accessible to the transit or receiving partners. The shipment remains in the pre-database until it is activated by a scan event, which may take up to 60 days before the data is removed if no scan occurs (Excluded are shipments with a DL or DI flag within the linehaul indicator field. These will be activated directly).

Until activation, the shipping partner can update any fields not printed on the label using a PUT request. Information that printed on the label is described in the field descriptions under https://api.eurodis.com/send-pan/documentation/index.html.

In case the label is requested via API, the activation process remains unchanged. The shipment will still be activated as described above, even when the label is requested.

If the existing ED collo barcodes are not included in the PUT request for updating, they will be deleted from the database. EURODIS will then create new ED collo barcodes, which will be provided as a response to the updating shipping partner.

This allows for the generation of new labels and barcodes, which can be used to correct any errors and ensure accurate tracking.

Additionally, the PUT request can create and add new ED collo barcodes alongside the existing ones. To do this, provide the existing collo barcodes and their associated information, while leaving the ED collo barcode field for the new parcels or pallets blank.

Example:


```
{
  "collis": [
    {
      "barcode": "12345678901234567890000",
      "identifierPalletOrCollo": "C",
      "weightOfUnit": 5.11,
      "boxType": "8",
      "boxDimensionCategory": "A",
      "boxId": "99887766551",
      "referencedEuroplateBarcode": null,
      "volumetric": null,
      "consignorsColloBarcode": null,
      "length": 43,
      "width": 29,
      "height": 32,
      "volumetricWeight": null
    }
    {
      "barcode": "",
      "identifierPalletOrCollo": "C",
      "weightOfUnit": 2.10,
      "boxType": "3",
      "boxDimensionCategory": "A",
      "boxId": "99887766555",
```

```
    "referencedEuroplateBarcode": null,
    "volumetric": null,
    "consignorsColloBarcode": null,
    "length": 32,
    "width": 21,
    "height": 29,
    "volumetricWeight": null
  }
```


The response from EURODIS will include a new ED collo barcode.

Please note that the example above only illustrates the barcode fields of the request. The PUT request must include all relevant shipment data fields according to the description.

5.2.3 Receive Shipment

You can retrieve all shipment data that has not yet been collected up to the time of your request by using the endpoint:



The response will provide details on the activated shipment data specifically for the requesting partner. This includes all relevant shipment information that has been processed and is ready for collection but has not yet been picked up at the time of the request.

5.3 Tracking Exchange

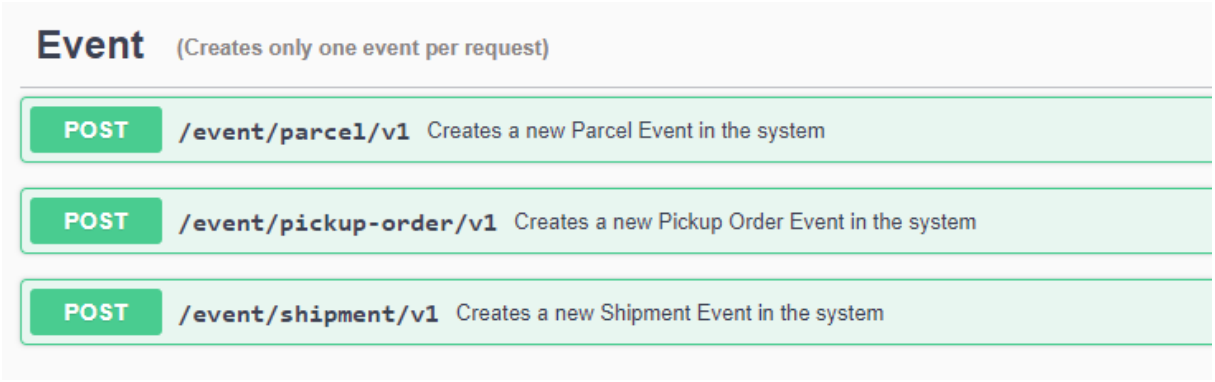
This chapter describes the exchange of tracking data via API within the EURODIS network.

5.3.1 Send Tracking

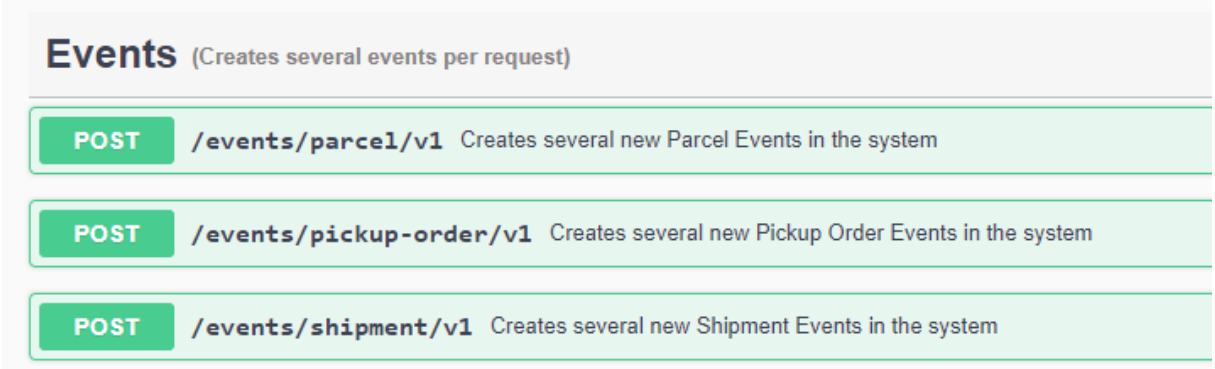
To send the tracking information of a shipment, follow the field descriptions provided at https://api.eurodis.com/send-tracking/documentation/index.html and ensure that the tracking information is accurately entered into the defined fields.

The shipping partner has the option to transmit events either individually or in bulk.

For individual event submission, use the "Event" section. This section is designed for sending details about a single event at a time:



For bulk event submission, use the "Events" section. This allows for the transmission of multiple events in a single request, which can be more efficient for processing large volumes of data.



EURODIS further classifies scan events into three distinct categories:

1. Shipment Level: Events related to the entire shipment. These events provide updates on the overall status and progress of the shipment as a whole.

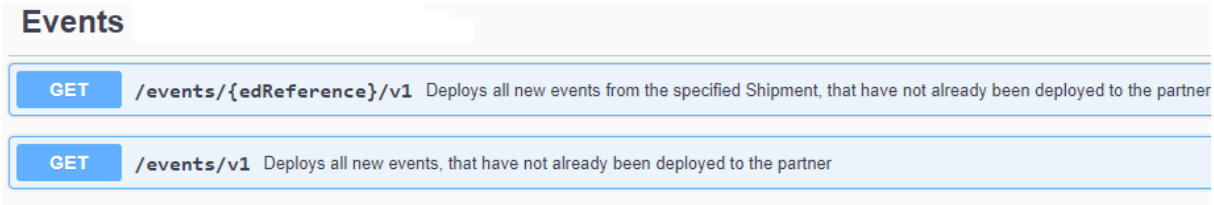
2. Collo Level: Events specific to individual collo (packages) within a shipment. These events track the status and progress of each collo separately, offering detailed insights into the handling and location of individual packages.

3. Pickup Level: Events pertaining to the pickup order. These include updates related to the scheduling and execution of the pickup of shipments from the execution partner.

Each category helps in accurately tracking and managing shipments, ensuring that all relevant information is captured and communicated effectively throughout the logistics process.

5.3.2 Receive Tracking

To track shipments using the EURODIS network, you can utilize the endpoint detailed in https://api.eurodis.com/receive-tracking/documentation/index.html



The requesting partner has the option to either query specific shipments or initiate a request to receive scan events that have not yet been collected up to the time of the request.

This allows the partner to either focus on particular shipments or obtain all relevant scan events that are still pending collection.

5.4 POD (Proof of delivery)

This chapter describes the proper exchange of PODs (Proof of Delivery) within the EURODIS network.

For detailed field descriptions related to exporting PODs, refer to the main document at https://api.eurodis.com/send-pod/documentation/index.html.

For information on importing PODs, please consult the documentation at https://api.eurodis.com/receive-pod/documentation/index.html.

The accepted formats for PODs include JPG, JPEG, PNG, TIF, TIFF, and GIF.

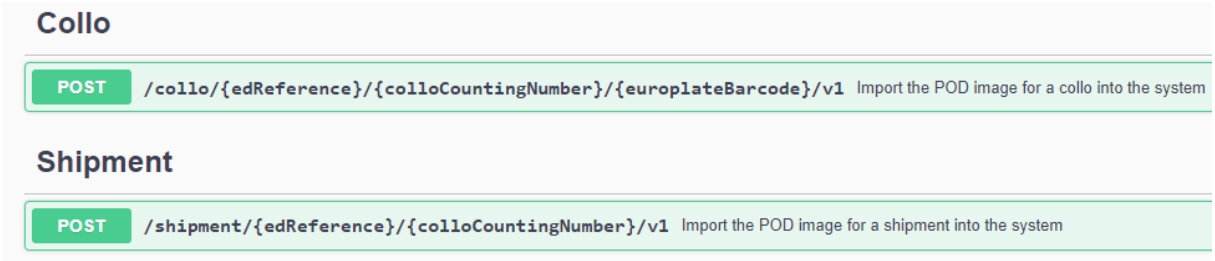
When EURODIS receives a POD from a delivery partner, EURODIS triggers the Scan Event 903, called "POD Available," to the dispatch partner. This Scan Event is not displayed in the Track & Trace system but serves as a trigger for the dispatch partner to request the POD.

The Scan Event 903 notifies the partner that the POD is ready for retrieval. The POD will only be made available for download via individual or bulk requests after the Scan Event 903 has been processed and acknowledged by the partner. This approach ensures that partners are promptly informed when a POD is available and can request it efficiently.

5.4.1 Send POD

The partner submits the PODs (Proof of Delivery) with a request. The partner is free to submit as many PODs as desired, no restrictions will be imposed. EURODIS stores the PODs in a separate database and generate links that direct to the PODs.

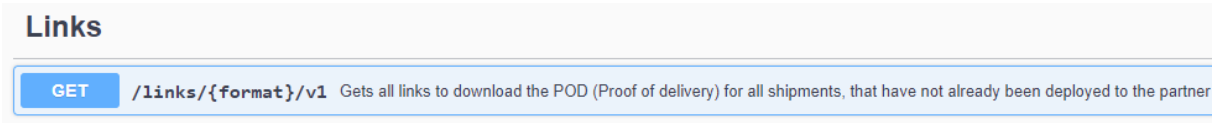
The partner has the option to submit PODs either at the shipment level or at the package level. To accommodate this, EURODIS has established two endpoints:



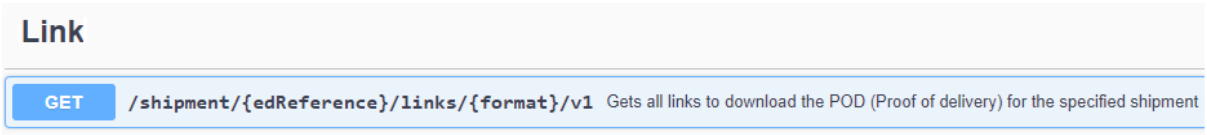
5.4.2 Receive POD

The partner has the option to request all PODs that have not yet been provided since their last request. The partner is allowed to make such requests no more frequently than every 30 minutes. If a request is made within 28 minutes of the previous request, an error message will be triggered, and the request will be rejected.

As a response, the partner will receive links to all available PODs. The partner has the option to either download the PODs directly from the provided links or simply save the links for future access:



It is also possible to request PODs for a specific shipment using the ED shipment number. When a partner makes such a request, EURODIS will provide all PODs associated with that particular shipment. The response will include each POD that EURODIS has received for the shipment identified by the ED shipment number:



5.4.3 Miscellaneous

- The generated link for responses allows a maximum of 3 downloads.
- The POD will be provided to the delivery partner only if one of the following scan events is present: '100', '101', '104', '416', '1100', '1104', '1416'. If a partner requests the PODs for a shipment without one of these Scan Events being available, an error message will be triggered.
- PODs are available in the EURODIS system for 13 months.
- In cases of partial delivery, PODs are not held back but are shared with the dispatch partner if a corresponding Scan Event is present.

- PODs will initially be available for a period of 3 months for individual requests. This duration may be subject to change in the future.

5.5 PUDO

This chapter describes the proper exchange of PUDO stations (Pick-Up and Drop-Off) within the EURODIS network.

5.5.1 PUDO Station

Each partner within the network has the opportunity to offer the PUDO stations from their domestic network as international PUDO stations, making them accessible for use by other partners across different countries.

To facilitate this, EURODIS provides a centralized database. In this database, each partner can enter address details of their PUDO stations, along with additional information such as whether Cash on Delivery (COD) is available, the maximum number of parcels that can be handled, and other relevant details.

This database can then be downloaded by each partner via an API request, allowing them to integrate this information into their own system.

This process is exclusively accessible through the API.

5.5.2 Enter PUDO Station in database

The relevant field descriptions for the API can be found at https://api.eurodis.com/pudo/documentation/index.html#/.

There are three key logics to highlight.

- Complete Data Submission:
  EURODIS expects that with each POST request, all currently available PUDO stations are provided. The update process is not incremental; instead, the latest POST request completely replaces all previous PUDO information for that partner.

- Time-Restricted Updates:
  Parcel stations can only be added to the EURODIS system during designated time windows. This restriction ensures that updates and data processing occur at specific intervals, likely to ensure data integrity. The allowed time frames for adding parcel stations are as follows (all times are in GMT+1):

  01:30 – 02:30
  09:30 – 10:00
  17:00 – 17:30

- Error Handling Based on Data Quality:
  Below 20% Corrupt Data: If the amount of corrupt data is below 20% (e.g., 8 out of 100 PUDO locations have corrupt data = 8%), only the corrupt PUDO locations will be rejected, and error codes will be triggered for those entries.
  20% or More Corrupt Data: If 20% or more of the data is corrupt (e.g., 21 out of 100 PUDO locations have corrupt data = 21%), 100% of the submitted data will be rejected, and error codes will be generated**.**

5.5.3 Download PUDO Station in database

While the database is being updated by partners, it is not possible to extract data from the database. The times during which parcel stations can be read out from the database are as follows (all times are in GMT+1):

00:00 – 01:30
02:30 – 09:30
10:00 – 17:00
17:30 – 00:00

During these periods, you can access and extract data from the database. Updates to the database are not allowed during these times to ensure the accuracy and consistency of the data.

5.6 Get Label

The following chapter describes how to request the EURODIS combined label via the API. The purpose of this label is to include the barcodes of the dispatch partner, the delivery partner and the standard EURODIS barcode (24 digit, 2of5 interleaved). The standard EURODIS barcode serves as the transit barcode and is mandatory for processing by all transit partners within the EURODIS network. You have the option to receive two different file formats, PDF or ZPL.

The link is for the programmer definition: https://testapi.eurodis.com/get-label/documentation

5.6.1 Request

First, the standard API request "create shipment" must be executed (the detailed description can be find in the chapter beginning of this document). As a response, you receive a collo barcode for each unit in the shipment and an ED shipment number. Together with the type of response, this is used as the basis for the label request.

The endpoint for single unit label request looks as follows:

GET `/collo/{barcode}/{type}/v1` Returns the label of the collo based on the transferred EURODIS collo (EUROPLATE), dispatch or delivery partner barcode

The endpoint for the labels of all the units within a shipment looks as follows:

GET `/shipment/{edReference}/{type}/v1` Returns all labels for the shipment based on the transferred ED reference

For detailed information please follow the link: https://testapi.eurodis.com/get-label/documentation

5.6.2 Response

The response is returned in the selected format (type). In the case of PDF, the file is provided as a Base64-encoded string. For ZPL, the raw ZPL command is returned. Along with the label, the response also includes two additional barcodes: Dispatch Partner and Delivery Partner, both provided as numeric values.


```
{
  "edReference": 7349706635114,
  "europlateBarcode": "560816153210285093333780",
  "deliveryPartnerBarcode": "JVGTCRDS0000000000002",
  "dispatchPartnerBarcode": "333000000250591361000000001001",
  "data": "JVBERi0xLjQNCiWio4+TDQozIDAgb2JqDQo8PC9UeXBlIC9YT2Jq
RGVjb2RlIFswIDEgMCAxIDAgMV0gL0ludGVycG9sYXRlIGZhbHNlIC9GaWx0ZXI
AAD5REAABAAAAAQEAAABREQAEAAAAAQAADsJREgAEAAAAAQAADsIAAAAAAAGGoA
```

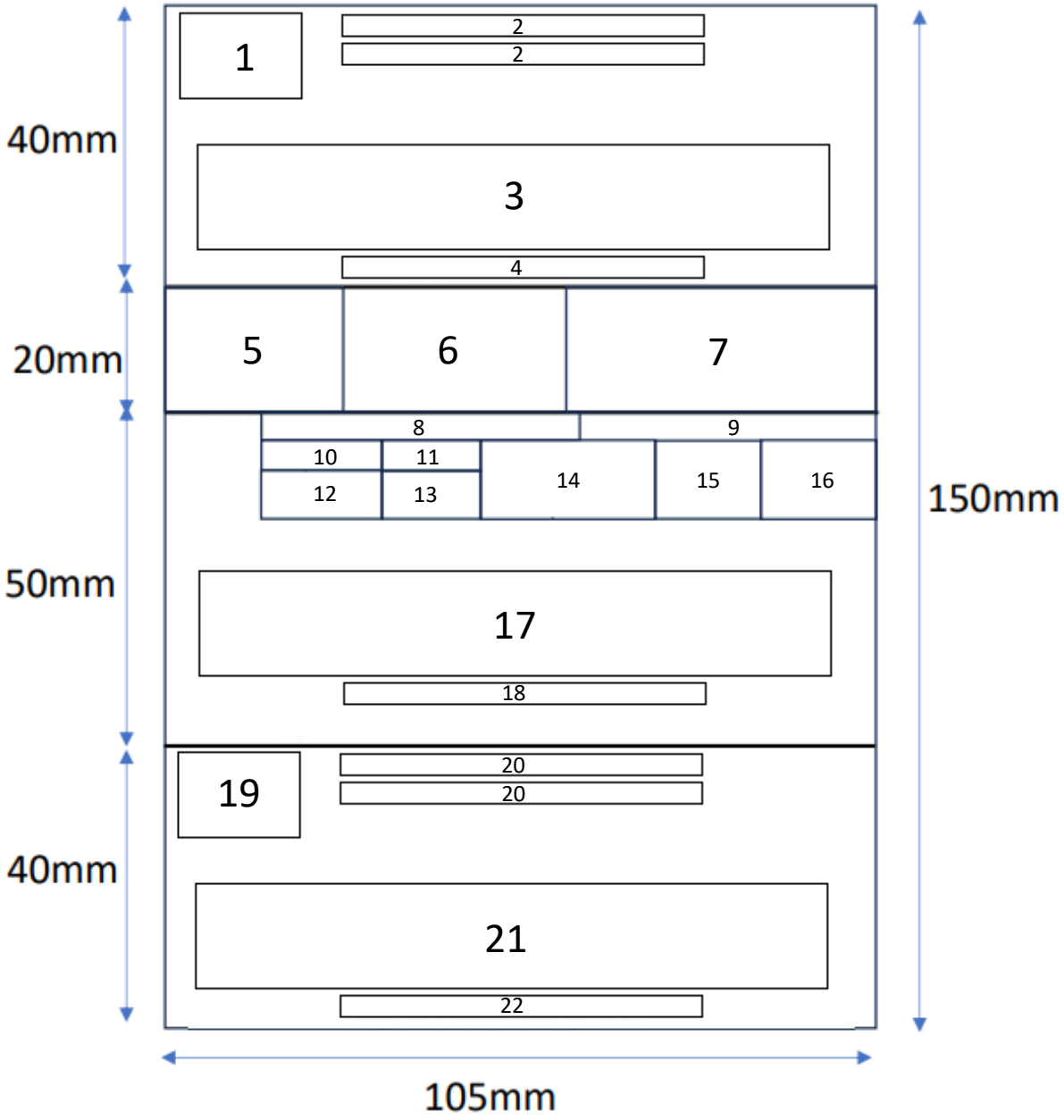
5.6.3 Label

Each partner receives a 40mm x 105mm container, which can be freely designed in cooperation with EURODIS. An API label contains three different containers. One for the dispatch partner, one for the delivery partner and one for the transit partner. The container for the transit partner always includes the standard EURODIS barcode in 2of5 interleaved format with 24 digits. In addition to the barcode containers, the label also provides general information about the shipment. A detailed description of each field is provided in the following chapter.

An example label here:



5.6.4 Label description



Field number	Description	Partner
1	Logo	Container of the Dispatch Partner
2	Free field, which can be used	
3	Barcode	
4	Barcode number	
5	Shipper name and shipper address	Shipment and unit information
6	Free field, which can be used	
7	Consignee name and consignee address	
8	Eurodis shipment number	
9	Unit reference	
10	PAN date	
11	Free field, which can be used	
12	Unit weight	
13	Number of unit in shipment	
14	Free field, which can be used	
15	Consignee country	
16	Consignee zip code	
17	Barcode	Container of the Transit Partner
18	Barcode number	
19	Logo	Container of the Delivery Partner
20	Free field, which can be used	
21	Barcode	
22	Barcode number	

5.6.5 Routing code

The routing code of the delivery partners is always generated centrally by EURODIS and then transmitted to the dispatch partner via the label. To enable central generation, the routing logic must be provided by the respective partner and implemented by EURODIS. The routing code can be placed either in the dedicated routing code field or within the delivery partner’s container. Additionally, the routing code will be included in the receive (import) data provided to the delivery partner.

6. Error codes PAN (shipment data)

The following table describes the system's response to missing information. EURODIS does not validate the content of the fields. It only checks whether the field is filled and if the type has been observed (Numeric, Alphabetic, Alphanumeric).

error code (FTP)	field name in API interface	error type message	warning message	error message (reject)	comment
20030	Line haul indicator	Line haul indicator not defined		x	
20050	Referenced EURODIS shipment number / Pickup order ID	No Original EURODIS Shipment reference supplied for a return shipment or Pick up order.		x	
20060	clearance timestamp	No clearance date or not in right format	x		
20070	entrance timestamp	Entrance date not in right format			
20090	Number of handling units	Number of handling units is 0	x		Also warning: If the number does not match the number of packages/pallets
20100	Shipment weight	Shipment weight is not defined or 0	x		
20160	terms of delivery (Eurodis only)	Invalid term of delivery			
29020		Error code has to be provided blank from partner			
30040	Contact telephone	Contact telephone must be filled		x	Reject If missing for: RT20-020 = A; RT30-020 = P
30060	Contact sms	Contact SMS must be filled			
30070	Contact E-Mail	Contact E-Mail must be filled			
30080	Company name	Company name field is mandatory		x	
30110	Street	Street is mandatory		x	
30140	City	City of delivery address unknown		x	
30150	Zip code	Invalid Zip code		x	
30160	Country code	Country code is unknown		x	
60020	Delivery Service	Invalid delivery service code			
60030	Delivery type	Invalid delivery type code		x	If delivery service RT60-020 is provided but RT60-030 is missing, then reject.
60060	Delivery time window start	Invalid time			
60070	Delivery time window end	Delivery window end date / time is earlier			

		than delivery window start date / time			
60080	Delivery date window start	Delivery window start date is either prior to request date or too far into the future			
60090	Delivery date window end	Delivery window end date is either prior to request date or too far into the future			
61020	IBAN	Invalid European IBAN (detailed checks tbd!)			
61030	BIC	Invalid European BIC (detailed checks tbd!)			
61040	Accountholder	No account holder supplied		x	
61070	Amount	Invalid or no amount		x	If missing, reject the COD shipment
61080	Currency	Invalid or no currency		x	If missing, reject the COD shipment
61090	Identifier for COD collection	Identifier for COD collection is <> 3 and <> 5		x	If missing, reject the COD shipment
62030	Pick-up date	Collection date is either prior to request date or too far into the future			
62040	Pick-up time - pickup ready	Collection time - latest time' is prior to 'Collection time - pickup ready'			
62050	Pick-up time - latest time	Collection time - latest time' is prior to 'Collection time - pickup ready'			
62090	Pick-up reference	Missing content		x	If missing, reject pickup request
62100	Number of coli	Missing number of colli		x	If all 4 are missing, reject pickup request
62110	Number of colli+	Missing number of colli +		x	If all 4 are missing, reject pickup request
62120	Number of pallets	Missing number of pallets		x	If all 4 are missing, reject pickup request
62130	Number of pallets+	Missing number of pallets		x	If all 4 are missing, reject pickup request
62150	Weight	Incorrect value or missing content		x	If missing, reject pickup request
62170	Currency for REF160	Invalid currency code			
63020	Commercial value of goods	Value of goods missing			

63030	Currency	Invalid or missing currency			
63040	Duty Code	Duty code missing			
63070	Description of Goods	Description of goods missing			
64020	Bill of Danger	Bill of danger missing			
64060	Tunnel limitation code	Invalid tunnel limitation code			
64070	Packing group	Invalid packing group			
64080	ADR-gross weight	ADR-gross weight missing	x		
64090	UN number	UN number missing		x	If missing, reject ADR shipment
64100	Transport category	Invalid transport category			
64110	Material description	Material description missing		x	If missing, reject ADR shipment
64130	Identifier mass or volume	Identifier mass or volume missing (neu hinzugefügt)		x	If missing, reject ADR shipment
65020	SMS pre advice	SMS pre advice		x	If delivery service requested, Contact SMS (consignee) as must be provided. Reject if missing.
65030	E-Mail pre advice	E-Mail pre advice		x	If delivery service requested, Contact E-Mail (consignee) must be provided. Reject if missing.
65040	Telephone pre advice	Telephone pre advice		x	If delivery service requested, Contact Telephone (consignee) must be provided. Reject if missing.
80020	Identifier pallet or collo	Invalid identifier pallet or collo		x	
80030	Europlate Barcode	Invalid barcode information			
80040	Weight of unit	Collo weight missing or 0			
80080	Referenced Europlate barcode	no referenced original Eurodis collo barcode		x	If missing, reject return shipment
80090	Volumetric	Volumetric data missing or 0			
80100	Consignor's collo barcode	Missing content in case of shipment type B, D, E		x	If missing, reject NB shipment
84030	Explanation of parcel type	Check if packaging is transmitted for ADR-goods			

7. Notes

7.1. Annex, templates

7.2. Further applicable documents

A03 Annex Data interchange & standard interfaces
Chapter 4 IT Requirements
https://api.eurodis.com/

7.3. References

7.4. Further explanatory notes, exemptions

8. Documentation

9. Change Management

10. Distribution list